

χ^2 Minimization II: Goodness-of-Fit

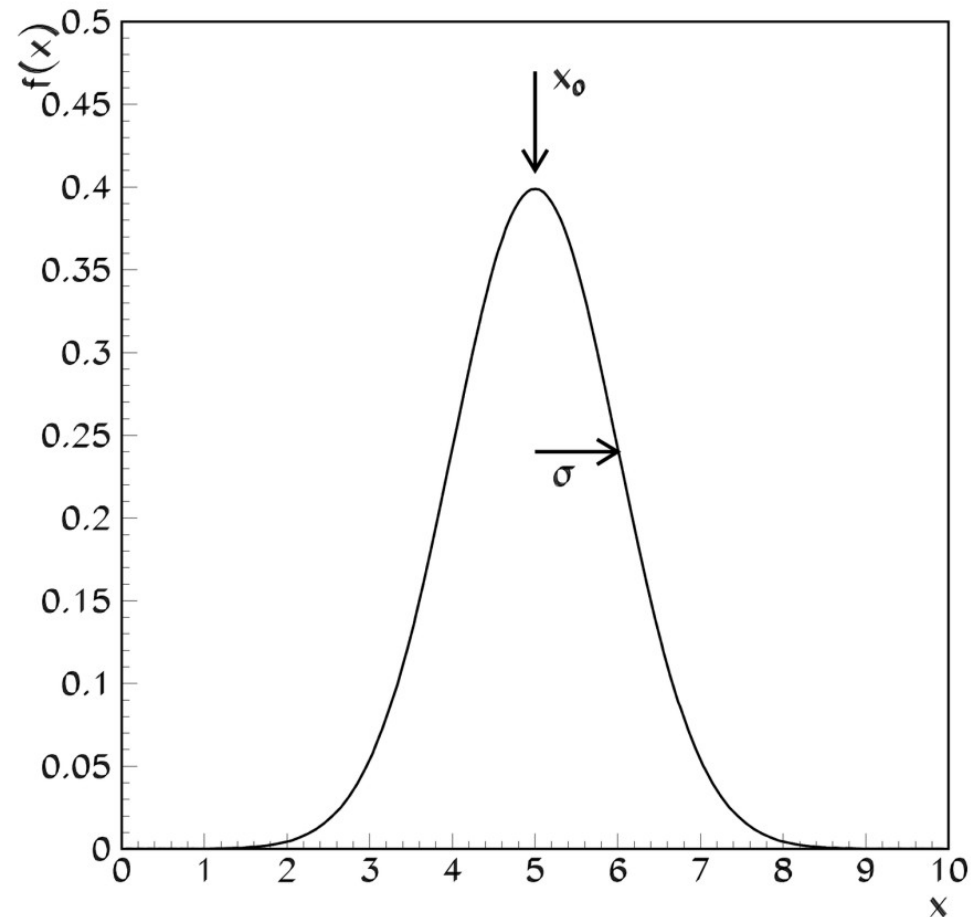
PHYS 4730, F2026

Statistical Inference

- **Estimating** model parameters from the data.
- Quantifying **confidence** in estimates.
- Quantifying **consistency** of data and model.

Gaussian (Normal) Distribution

- In many physical measurements, errors may be taken to be *normally distributed*.
- In these cases, the *Maximum Likelihood* technique reduces to the technique of *Chi-Squared Minimization*.
- Assessing goodness-of-fit in these cases is straightforward.



$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\bar{x})^2/2\sigma^2}$$

Evaluating Fits: χ^2

- Definition of χ^2

$$\chi^2 = \sum_{i=1}^n \frac{(f(x_i) - y_i)^2}{\sigma_{y_i}^2}$$

- Absolute magnitude of χ^2 only has meaning relative to the number of **degrees of freedom**
- **#DOF** = (# of data points) -
(# **parameters determined from the data**)
- **#DOF** = (# of data points) - (# **constraints**)
- $\chi^2 / \text{\#DOF} \sim 1$ for good fit (rule of thumb)
- What would “zero degrees of freedom” mean?

Goodness-of-Fit

Assertion: For N random variables x_i drawn from a normal distribution with fixed mean μ , standard deviation σ , the quantity

$$\xi = \sum_{i=1}^N \left(\frac{x_i - \mu}{\sigma} \right)^2$$

will be an IIDRV following a χ^2 distribution:

$$f(\xi|k) \equiv \chi^2(\xi|k) = \frac{1}{2^{k/2}\Gamma(k/2)} \xi^{k/2-1} e^{-\xi/2}$$

where $k = N$.

Assertion: If a function $f(x|\theta)$ is fit to a set of points y_i by χ^2 minimization, then the quantity

$$\xi = \sum_{i=1}^N \left(\frac{y_i - f(x_i)}{\sigma_i} \right)^2$$

will be an IIDRV following a χ^2 distribution with $k = N - N_p$ (N_p is the number of parameters determined from the data.)

Are the data and model consistent?

True:

1

Goodness-of-fit

False:

0

- “Goodness-of-fit”
 - aka “quality factor”
 - aka “p-value”
 - aka “chance probability”
- Number between $0 \rightarrow 1$
- Quantifies the probability of obtaining your χ^2 / N_{df} by chance.

Are the data and model consistent?

$$Q = \Gamma_{inc}^{up} \left(\frac{N_{df}}{2}, \frac{\chi^2}{2} \right)$$

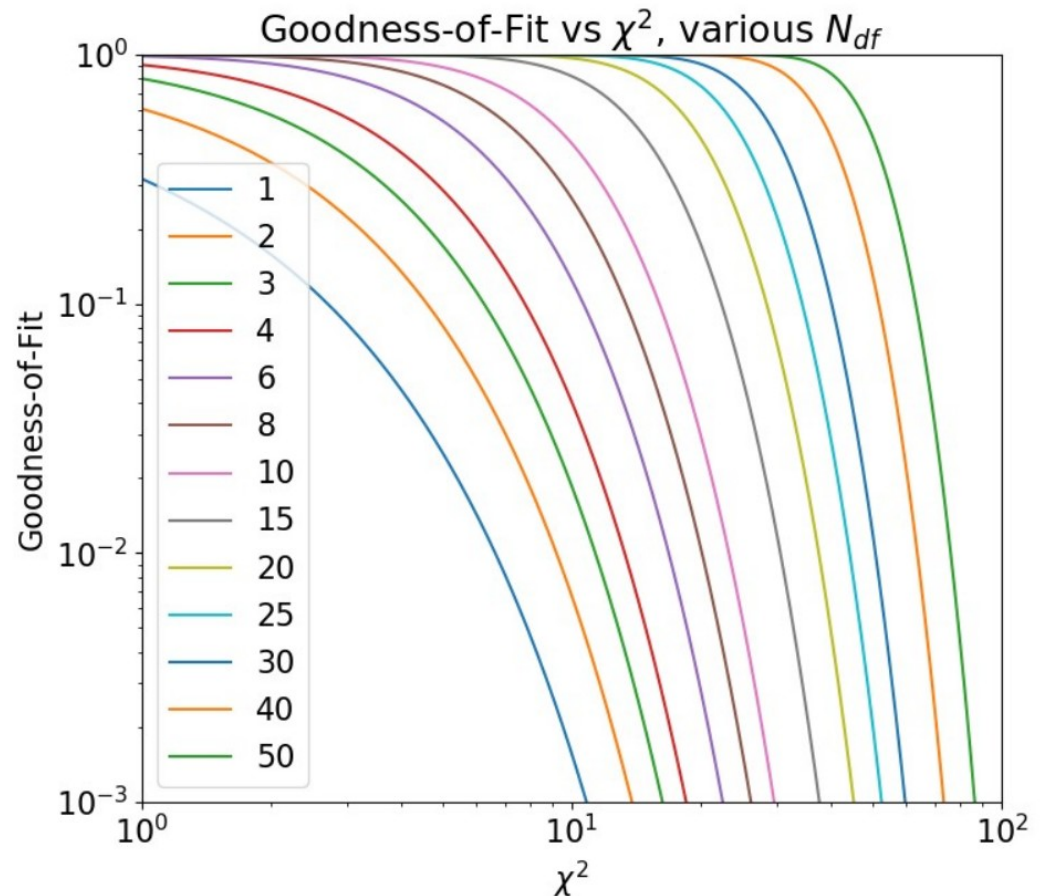
$$\Gamma_{inc}^{up}(a, x) = \frac{1}{\Gamma(a)} \int_x^{\infty} t^{a-1} e^{-t} dt$$

$$\Gamma(a) = \int_0^{\infty} t^{a-1} e^{-t} dt$$

Q = `scipy.special.gammaincc`

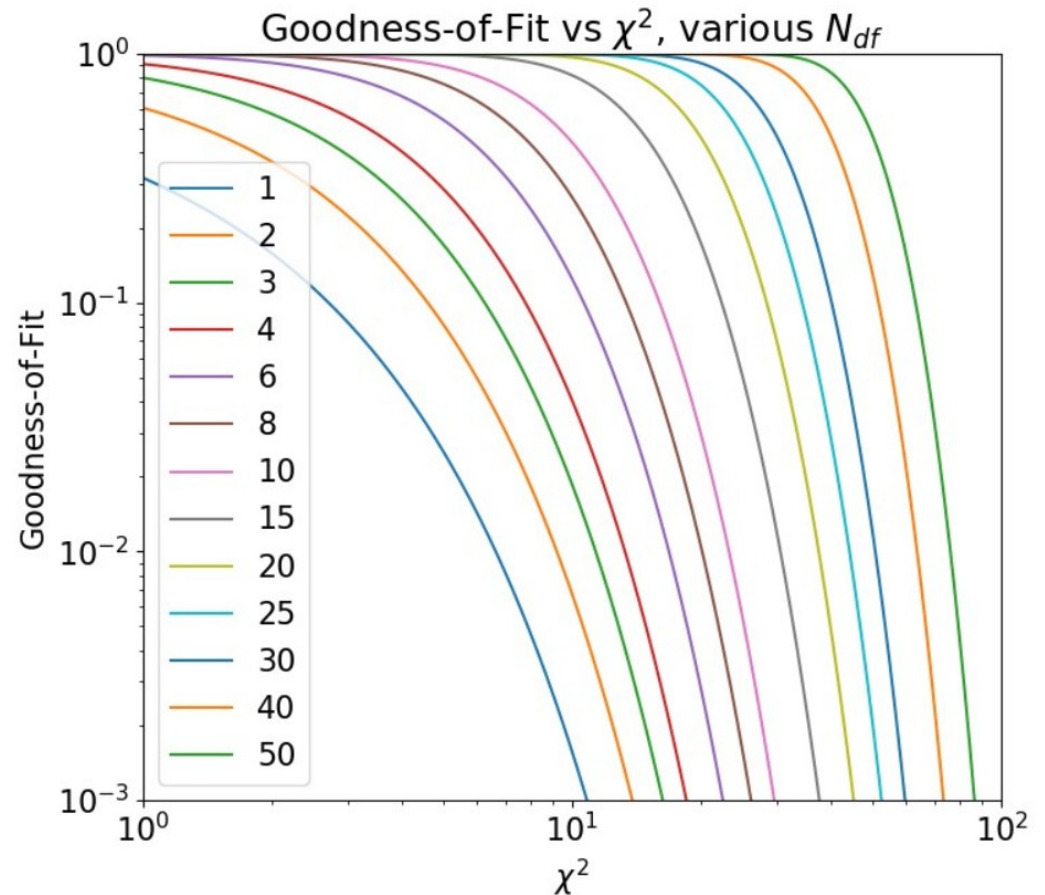
Are the data and model consistent?

- In a fit to experimental data, I obtain
 - $\chi^2 = 10.0$
 - $N_{df} = 6.0$
- Goodness of fit = 0.125
- “If the model were true, I’d expect the same χ^2 or larger in 12.5% of experiments identical to mine.”
- Approximate “two sigma” with $Q = 0.05$ in assignments.



What is the best-pick model?

- It is possible to get a better χ^2 by increasing the number of parameters fit.
- If the goodness-of-fit hasn't improved (gotten closer to 1), you haven't gained anything!
- **The best model is the one with the largest goodness of fit.**



Additional Reading:

Taylor: *An Introduction to Error Analysis*

Bevington: *Data Reduction and Error Analysis for the Physical Sciences*

Press *et al*, *Numerical Recipes*